

A novel perspective on negative bias in polar questions: The view from Russian

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Abstract In contrast to well-researched *wh*-questions, the semantics and pragmatics of polar, and especially biased, questions in Slavic languages have not received much attention within formal approaches to meaning (Šimík forth.). This chapter addresses this gap through a focused investigation of two Russian question particles, *razve* and *neuželi*, which, I argue, serve as a window on the nature of question bias. Previous literature characterizes *razve* and *neuželi* as markers of negative speaker bias and positive contextual bias, roughly corresponding to English *really* (Geist and Repp 2023; Repp and Geist 2025). Zooming in on subtle differences in their conventional meaning and conversational dynamics (cf. Baranov 1986; Bulygina and Shmelev 1997), I show that such characterization is inaccurate and use the present case study to highlight the limitations of current approaches to question bias as a whole. My goals are fourfold: (i) to provide an in-depth description of *razve* and *neuželi* in isolation, demonstrating that they do not fit the extant typology; (ii) to propose, building on Korotkova (2023), that negative speaker bias in questions tracks the belief revision potential of a situation, which explains the behavior of *razve* and *neuželi* in a principled way; (iii) to make a case for viewing contextual bias as an evidential notion; and (iv) to endorse the idea that discourse effects associated with question bias needn't be hard-wired to the semantics of expressions that give rise to those effects.

Keywords belief revision, discourse particles, evidence in language, polar interrogatives, question bias, Russian

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*Razve ty mne ne skažeš snova
Pobedivšee smert' slovo
I razgadku žizni moej?*
And then won't you say
One more time
That word that overcomes death
And unties the enigma of my life?

—*Poem Without a Hero*, Anna Akhmatova
(Translated by L. Mayhew & W. McNaughton)

1 Introduction

There has been a burgeoning interest in polar questions of all stripes within formal approaches to meaning. And yet polar questions in Slavic languages, especially compared to their well-researched *wh*-cousins (Citko 2024), have escaped systematic attention, and their semantics and pragmatics remain underexplored (Šimík forth.). In particular, biased questions—those conveying the speaker's attitude towards the likelihood of the prejacent—are surprisingly understudied across Slavic. This chapter addresses this empirical gap through a focused investigation of two Russian question particles, *razve* and *neuželi*. Zooming in on subtle differences in their conventional meaning and conversational dynamics, I argue that these particles serve as a window on the phenomenon of question bias.

In well-studied languages like English, bias in questions is conveyed by means not earmarked for this function, such as negation or intonation, which makes it difficult to disentangle semantics from pragmatics (Goodhue, this volume). Many other languages, however, have dedicated constructions that encode the speaker's attitude, including discourse particles. Russian is one such language and has a host of particles (Shimchuk and Shchur 1999; Zibatow 1990). I focus on *razve* and *neuželi* as, among various elements with bias-like effects, they are the only two limited to polar questions¹ and thus make a perfect case study. My goals are fourfold:

- to provide an in-depth description of *razve* and *neuželi* in isolation, demonstrating that they do not fit the extant typology;
- to propose, building on Korotkova (2023), that negative speaker bias in questions tracks the belief revision potential of a situation, which explains the behavior of *razve* and *neuželi* in a principled way;
- to make a case for viewing contextual bias as an evidential notion;
- to endorse the idea that discourse effects associated with question bias needn't be hard-wired.

Section 2 situates *razve* and *neuželi* within current typology of bias and outlines my proposal. Section 3 provides the background on polar interrogatives in Russian, including prosody. Section 4 is the empirical heart of the paper and discusses the particles in detail. Section 5 proposes an analysis of *razve* and *neuželi* as purely doxastic operators and shows how this semantics derives their conversational uses. Section 6 concludes.

¹*Razve* also introduces nominal and clausal exceptives, which I will treat as synchronically unrelated to its life in questions (see Zalizniak 2020 on the diachrony).

2 *Razve*, *neuželi* and the typology of bias

Questions containing *razve* and *neuželi* are often ascribed doubt, confusion, surprise or disbelief (Baranov 1986; Bulygina and Shmelev 1997; Rathmayr 1985:115-120,137-155; Restan 1972:437-464, Shvedova et al. 1980:388, Zalizniak 2020, a.o.). Each particle is compatible with various emotions: *razve*-Qs convey puzzlement (4b), indignation (18), resignation (19), while *neuželi*-Qs express curiosity (20), contemplation (21), wonder (23). As I will argue, *razve* and *neuželi* encode beliefs, and the speaker's psychological state is not part of semantic composition or pragmatic calculation.

Razve and *neuželi* are presumptuous, even rude, in neutral inquiries when the speaker does not signal their attitude towards the prejacent (this effect comes from particles rather than special syntax or intonation; Section 4.1).²

(1) *At a job interview:*

#**Razve/neuželi** vy prepodavali latyn?
RAZVE/NEUŽELI you:PL.NOM teach.PL.PST Latin.ACC.SG
 'Have you ever taught Latin? (I thought not.)'

Current research identifies two types of question bias (Goodhue, this volume): (i) **speaker bias**: the speaker's prior attitude towards the prejacent (e.g., belief or preference), and (ii) **contextual bias**: mutually available evidence bearing on the prejacent. Using Do-maneschi et al.'s (2017) paradigm, (2) manipulates different types and polarity of bias (positive, negative, neutral). *Razve* and *neuželi* are only licensed in Scenario C2: the speaker's prior belief that $\neg p$ conflicts with contextual evidence supporting p .

(2) A. **Neutral doxastic**: *I meet you for the first time, we go out for lunch.*

- # 1. **Neutral contextual**: I order for both us and check with you beforehand.
- # 2. **Positive contextual**: You order beetroot hummus.
- # 3. **Negative contextual**: You avoid all beet mezzes.

B. **Positive doxastic**: *I was sure you like beets.*

- # 1. **Neutral contextual**: I invite you over and double-check before cooking.
- # 2. **Positive contextual**: We go out for lunch, you order beetroot hummus.
- # 3. **Negative contextual**: We go out for lunch, you avoid all beet mezzes.

C. **Negative doxastic**: *I was sure you hate beets.*

- # 1. **Neutral contextual**: I invite you over and double-check before cooking.
- ✓2. **Positive contextual**: We go out for lunch, you order beetroot hummus.
- # 3. **Negative contextual**: We go out for lunch, you avoid all beet mezzes.

Razve/neuželi ty ljubiš' sveklu?
RAZVE/NEUŽELI you:SG.NOM love.2SG.PRS beet.ACC.SG
 ≈ 'Do you like beets?'

Razve and *neuželi* can express preferences in addition to beliefs, but only the doxastic, and not bouletic, component is hard-wired (cf. Bulygina and Shmelev 1997:274), unlike

²I use the scientific transliteration throughout. Glosses: 1,2,3 person, ACC accusative, CONV converb, DAT dative, GEN genitive, F feminine, INC inchoative, INF infinitive, INSTR instrumental, M masculine, N neuter, NEG negation, NOM nominative, PL plural, PRED predicative, PREP prepositional, PRES present, PTCL particle,PTCP participle, PST past, SG singular, SUBJ subjunctive. Pitch contours were generated using Praat.

with some other expressions of bias in, e.g., Czech (Šimík et al. 2024) or Farsi (Mohamadi, this volume). (3) illustrates. Neither particle is licensed just by a preference that $\neg p$ (Scenario A1) and both are compatible with preferences either way if the relevant belief is present (Scenarios B1-B3).³

(3) A. **Desire without belief: neutral doxastic, positive contextual**

#1. **Negative bouletic:** I genuinely have no idea whether smoking is allowed in this bar we’re about to go to, since countries vary wildly, but as a non-smoker, I strongly prefer that it isn’t. As we enter, I notice somebody smoking a joint.

B. **Belief with our without desire: Negative doxastic, positive contextual**

We’re in what we thought was a non-smoking bar. Someone lights a cigarette.

✓1. **Positive bouletic:** An avid smoker, I am delighted to be mistaken.

✓2. **Negative bouletic:** An adamant non-smoker, I am dismayed to be mistaken.

✓3. **Neutral bouletic:** An alien explorer, I could care less.

Razve/neuželi zdes’ možno kurit’?

RAZVE/NEUŽELI here can.PRED smoke.INF

‘Can one smoke here?’

The data above confirm the intuitions expressed, but never explicitly checked, in previous work. In line with current views, Ljudmila Geist and Sophie Repp, henceforth G&R, suggest that *razve* and *neuželi* encode negative speaker bias and positive contextual bias, reminiscent of English *really* (Geist and Repp 2023; Repp and Geist 2025). I argue that this descriptive characterization is incomplete⁴ and that the existing accounts do not capture the behavior of the particles.

Empirically, I show that the resemblance to *really* is superficial. *Really* always conveys a strong bias against p at the moment of conversation (Romero and Han 2004; Bill and Koev 2022). *Razve* conveys such attitude only in a subset of cases (Section 4.2), while *neuželi* conveys that the speaker considered p unlikely but now is (nearly) certain that p (Section 4.3). Furthermore, *razve* and *neuželi* impose distinct evidential requirements (Section 4.4), and it’s unclear whether *really* has any. Additional insight comes from literary translations: while a detailed corpus study is outside the scope of this paper, it’s clear that only some instances of the Russian particles correspond to *really* in the parallel section of the Russian National Corpus, which shows that *really* is not their English equivalent (pace G&R).

Conceptually, I use the present case study to highlight the limitations of current approaches to bias as a whole. First, the literature often construes ‘speaker bias’ as a static notion: the speaker’s belief acquired prior to, and possibly held during, the conversation (though see Heim and Wiltschko 2022 on the timing of bias, Csipak and Zobel 2014 on belief change as a licensing condition for German *etwa*, and Kiss 2024⁵ on the role of belief revision in the semantics of declarative questions across languages). Second, the literature largely treats ‘contextual bias’ as a homogeneous notion: evidence available to interlocutors. It’s difficult to characterize the particles using these concepts. They either should

³I thank Robert van Rooij for raising this issue, and Felix Frühauf for discussing relevant examples with me.

⁴G&R focus on negative questions, including with the particles, and don’t aim at a complete formal account. My criticism explores a version of their ideas.

⁵I thank Hans-Martin Gärtner for bringing this important work to my attention, though due to timing I cannot do it full justice here.

occupy the same cell in the typology as markers of negative speaker bias and positive contextual bias, which doesn't capture their differences, or two distinct cells, which doesn't capture their similarities. Something clearly goes amiss.

First, I argue that *razve* and *neuželi* require a dynamic approach that differentiates prior vs. current belief and propose that negative speaker bias occurs in situations with belief revision potential, the speaker induced to possibly change their beliefs in light of new evidence.⁶ Second, I show that *razve* belongs to a group of expressions sensitive to the direction of reasoning and propose to treat contextual bias as an evidential notion. Building on [Korotkova \(2023\)](#), I argue that *razve*-Qs and *neuželi*-Qs both require low prior credence in the prejacent and the presence of contextual evidence supporting it (as previous research suggests), but differ in evidential requirements (Section 4.4) and the posterior credence: with *razve*-Qs, the speaker is often deliberating, while with *neuželi*-Qs, they are on the verge of acceptance. (4) illustrates.

(4) *The author to his brother, who wrote about leaving Kyiv for Odesa.*

- a. **Neuželi** ty [...] poedeš' v Odessu,
NEUŽELI you:SG.NOM go.2SG.PRES in Odesa.PREP
 ≈'I can hardly believe you are going to Odesa'
- b. **razve** nel'zja obojti čtenie lekcij na
RAZVE impossible.PRED avoid.INF reading.ACC.SG lecture.GEN.PL on
 etot semestr?
 this.M.ACC.SG semester.ACC.SG
 ≈'Is it impossible to get out of teaching this semester? (I thought not).'
- (Letters to Alexander Kovalevsky, Vladimir Kovalevsky)

Neuželi (4a) conveys prior belief that the addressee wouldn't leave for Odesa. *Razve* (4b) conveys prior belief that the addressee wouldn't have to teach. Each situation carries belief revision potential, as each prior belief is at odds with contextual evidence: the addressee is clearly leaving, probably because they have to teach. The key difference between the two questions is in the speaker's posterior belief and, as a result, in their communicative goals. The *neuželi*-Q signals near-certainty and requests a confirmation, while the *razve*-Q signals genuine ignorance and requests an answer.

In Section 5, I argue that each particle conventionally encodes only the conditions that may bring belief revision about: (i) the speaker's initial low credence in the prejacent, (ii) the speaker's current credence ($\neq$ initial belief): low to undecided with *razve*, high to full with *neuželi*, (iii) contextual evidence: both particles require at least good-enough evidence supporting the prejacent, but only *razve* requires an inference from effect to explanation. I further show that this semantics determines: (i) the subjective strength of contextual evidence, and (ii) the range of conversational moves each particle performs.

3 Polar interrogatives

Setting alternative questions aside, Russian has two types of polar questions ([Comrie 1984](#); [Restan 1972](#); [Schwabe 2004](#); [Shatunovskij 2016](#):135–180; [Shvedova et al. 1980](#):387):

1. *li*-Qs: questions with the focus particle *li*;

⁶I formalize this intuition using probabilistic updates, but nothing hinges on this. If the idea is right, I expect future research to adjudicate between different models of belief revision (cf. [Hansson 2022](#); [Rott 2021](#)) based on their suitability for question bias. I thank an anonymous reviewer for bringing this up.

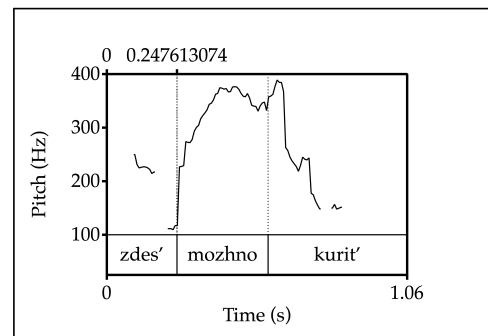
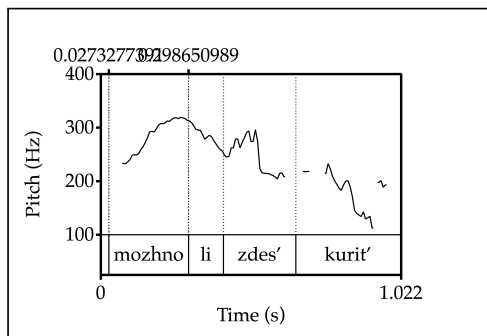
2. **Decl-Qs**: declarative strings with the ‘Q-peak’ intonation (term from Esipova 2024).

Li-Qs are more formal (but not archaic, pace Munteanu and Kiss 2025; Schwabe 2004). While there are pragmatic nuances between the two strategies (Esipova and Korotkova 2023; Shatunovskij 2016), neither encodes bias.⁷ Both *li*-Qs and Decl-Qs abound in unbiased contexts where the speaker has no expectations or evidence about the answer (5), and *li*-Qs are the only way to ask such questions in written official communication.

(5) *Visiting a new city, I’d like to know the local rules—I genuinely have no idea.*

a. Možno **li** zdes’ kurit’?
can.PRED LI here smoke.INF
‘Can one smoke here?’

b. Zdes’ možno kurit’?
Here can.PRED smoke.INF
‘Can one smoke here?’



3.1. *Li*-questions

Li is a second-position clitic (Bošković 2001; Franks and King 2000). Its host is the the focus of the question or a part of the focus-bearing constituent, which is always fronted. *Li*-Qs have no final rise and the focused element has the same prosodic profile as ordinary focus marking (Meyer and Mleinek 2006; Meyer 2008; Yanko 2019).

Because only *li*-Qs occur in embedded polar (≠alternative) questions (6), *li* has been analyzed as an interrogative complementizer (Brown and Franks 1995; Franks and King 2000; Grebenyova 2007; Kazenin 2006; Schwabe 2004). I review one argument for the low attachment of *li* (see Bošković 2001; Rudnitskaya 2000; Shushurin 2024): *li*-phrases are retained in sluicing (7), a special type of ellipsis under question-embedding predicates (Merchant 2006).⁸

(6) Masha sprosila/uznala/somnevaetsja, tancuet ***(li)**
Masha.NOM ask.F.SG.PST/learn.F.SG.PST/doubt.3SG.PRES dance.3SG.PRES **LI**
Tata.
Tata.NOM
‘Masha asked/learned/doubts if Tata danced.’

(7) Nina zavtra uezžæt, no ja ne znaju, nadolgo **li**
Nina.NOM tomorrow leave.3SG.PRES but I.NOM NEG know.1SG.PRES for.long **LI**
[Nina zavtra uezžæt].
Nina.NOM tomorrow leave.3SG.PRES
‘Nina is leaving tomorrow, but I don’t know if she is leaving for long.’

(adapted from Shushurin 2024: ex.21a)

⁷Both *li*-Qs and Decl-Qs are incompatible with negative contextual bias for *p* when they have a positive preajcent. Current research suggests that this is a universal property of positive polar questions that can be derived pragmatically (Goodhue 2022b; van Rooij and Šafářová 2003).

⁸Erschler (2017) demonstrates that this is sluicing vs. other type of ellipsis.

Interrogative complementizers are typically ungrammatical in sluices (Merchant 2001), so the data suggest that *li* is not base-generated in C. The exact analysis of non-*wh*-sluicing is unimportant here: *li*-phrases can be treated as moving to SpecCP (Shushurin 2024) or as staying in a low left-peripheral position, such as focus (in line with Rudnitskaya 2000; see Marušič et al. 2018; Craenenbroeck and Lipták 2013 on focus remnants of sluicing in Slovene and Hungarian, respectively). Crucially, *li* does not behave as a clause-typing element. Additional support for the non-complementizer view comes from *li*'s non-interrogative uses.

Li is often seen outside of questions, including in sentential epistemic adverbials *edva li*, *vrjad li* 'unlikely' and *čut' li* 'hardly' ('barely' + *li*), in the epistemic particle *čto li* ('what/something' + *li*; Bernasconi 2023; Onoeva 2024), and the alternating disjunction *to li X, to li Y* (Ivlieva 2024). Historically, *li* participated in the standard disjunction *ili* (< *i* 'and' + *li*) and the conditional conjunction *esli* 'if' (< *est* 'be.3SG.PRES' + *li*). There is a deep semantic connection between polar questions and disjunction (Mascarenhas 2009 and later work within Inquisitive Semantics), and disjunctions often are diachronic sources of question particles (Walkden et al. 2025). Furthermore, multiple languages have elements that systematically appear in polar questions, disjunctions, conditional antecedents and indefinite pronouns (Szabolcsi 2015). These facts call for a unified analysis of *li* across the board, even though *li*'s synchronic distribution is more constrained than that of the particles discussed by Szabolcsi. While I leave a compositional analysis for future research, I maintain that only a non-complementizer view on *li* (supported by independent syntactic evidence) is compatible with *li*'s other uses, since in those uses it definitely doesn't occupy C and can also occur in declaratives.

3.2. Declarative string questions

Declarative string questions (5b) have the same surface word order as declarative string assertions (SVO with permutations possible, e.g., due to information structure and other factors; Bailyn 2011; Griбанова 2017; Jasinskaja and Šimík forth.), but are reliably distinguished from assertions in perception and production through intonation (Makarova 2007; Rathcke 2006). Neutral, out-of-the-blue assertions have broad sentential focus: early f0 alignment with the onset of the accented vowel, gradual low rise (assertions can also have narrow focus). Decl-Qs feature the 'Q-peak' (term from Esipova 2024), a special prominence on the semantically focused constituent: late f0 alignment, steep high rise. Note that Russian is typologically unusual: questions do not have a rising intonation. The Q-peak differs from ordinary focus marking that occurs in, e.g., assertions, *li*-Qs or *wh*-questions in shape and height (Meyer and Mleinek 2006), and one of its main functions is to request a reaction from the addressee (Esipova 2024; also Shatunovskij 2016:143).

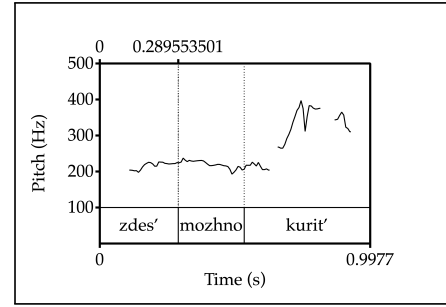
Q-peak placement gives rise to distinct discourse effects (observed in Restan 1972:155-160 and Shatunovskij 2016:156-162; see detailed discussion in Esipova and Romero 2023). Q-peak on the main predicate (or a different constituent): neutral question (5b), nearly synonymous to a *li*-Q with the fronted main predicate. Clause-final Q-peak ($\neq$ rising tune, there is a rise and fall): the utterance requires the presence of a non-trivial higher-order QUD. Such 'explanation-seeking' questions (term from Esipova and Romero) often occur when there is some contextual evidence supporting the prejacent (8), and as such resemble questions with 'contextual bias'. Importantly, the clause-final contour in (8) is infelicitous in the neutral-inquiry context from (5), and the utterance from (5b), with the Q-peak on the main predicate, is infelicitous in the non-neutral context in (8).

(8) *I see people smoking on the train.*

Zdes' možno kurit'Q?

Here can.PRED smoke.INF

'Can one smoke here?'



Summing up, Russian has two strategies for neutral polar questions: *li*-Qs, with the focus particle *li*, and Decl-Qs, with declarative strings. Each strategy has a distinct prosodic and syntactic profile, and in many contexts, they are nearly interchangeable.

3.3. A note on negation

Negative questions feature prominently in the literature on bias—thus, the so-called high negation in English or German introduces positive speaker bias for the prejacent (*Don't you like beets?* conveys that I think you do). However, Russian facts are controversial due to (i) the understudied role of intonation and (ii) the lack of evidence for the ambiguity of high negation, both of which are important for a complete analysis.

Negative *li*-Qs and Decl-Qs may, but don't have to be, non-neutral and are often preferred in polite requests (Restan 1972:288–350; Shatunovskij 2016:162–176; Zanon 2025). Negation in Russian, including in polar questions, can merge low or high (Abels 2005; Brown and Franks 1995). For *li*-Q, Zanon (2025) shows that high negation is correlated with non-neutrality (without elaborating on the nature of bias).⁹ Shatunovskij (2016:173) notes in passing that these questions have a special intonation, so the source of bias is unclear. For Decl-Qs, negation has been assumed to encode bias (G&R, Onoeva and Šimík 2023), but these studies do not control for intonation, and are therefore insightful, but inconclusive.

Given that Decl-Qs allow a non-neutral explanation-seeking contour, the observed bias effects of negative Decl-Qs may come not from high negation (as in English), but from intonation or possibly even both. To fully understand this, we need at least four types of Decl-Qs stimuli (positive/negative prejacent, neutral/explanation-seeking contour) matched with nine bias conditions (Domanechi et al.'s (2017) paradigm; Section 2). Additionally, it is imperative to control for low vs. high negation (see Zanon for tests). At the moment of writing, such studies do not exist and the scrupulous work on the interaction of bias, negation and intonation in Russian is in its nascent stages (see, e.g., Onoeva and Razguliaeva 2025).

Another issue is the purported ambiguity of high negation, the so-called outer (positive speaker bias that *p*) vs. inner negation (positive speaker bias that $\neg p$), first proposed for English by Ladd (1981) and widely used to motivate accounts of bias via dedicated illocutionary operators that arguably create the ambiguity:

- (9) a. [Q [$\neg p$]] [low negation]
 - b. [Q [$\neg Op_1$ [*p*]]] or [Q [Op_2 [*p*]]] [outer high negation]
 - c. [Q [Op_1 [$\neg p$]]] [inner high negation]
- (adapted from Goodhue, this volume)

⁹Shushurin (2024:11–13) incorrectly treats all negative *li*-Qs as biased.

G&R assume that Russian high negation in Decl-Qs is ambiguous, producing two distinct kinds of bias corresponding to the LFs in (9b) and (9c). This crucial assumption warrants further scrutiny, and I review one example here.

- (10) Julja **eščo** ne napisala tekst?
 Julja.NOM **yet** NEG write.3SG.PST text.ACC.SG
 ‘Has Julja not written the text yet? (=Is it not the case that Julja has already written the text?)’
 (adapted from Repp and Geist 2025:119,ex.13)

The outer-inner ambiguity can arguably be diagnosed by polarity items. Russian adverb *eščo* ‘yet/still’ requires negation with perfective verbs in the past tense (Paducheva 2013:241-247; 2014), and G&R assume that the adverb’s presence in cases like (10) is indicative of inner high negation, (9c). However, a viable alternative is undiscussed: the putative inner reading of high negation in (10) can be low (propositional) negation, (9a), which is precisely what Zanon proposes for the *li*-counterparts of (10).

According to G&R’s experimental data, (10) is acceptable in scenarios with the speaker’s prior belief that *p* (negative speaker bias for negative prejacent) and evidence supporting $\neg p$ (positive contextual bias for negative prejacent), but that alone does not prove the presence of high negation or a bias-inducing operator at LF, as low negation questions (*Has Julia not written the text yet?*) are also acceptable, though dispreferred, in such scenarios (Domaneschi et al. 2017). Furthermore, (10) is compatible with different intonational contours, which also may play a role in the licensing of negation-sensitive expressions (a fact not discussed by G&R).

The very existence of the outer-inner ambiguity has been challenged. Based on a variety of semantic diagnostics, including presupposition projection, Goodhue (2022b, this volume) establishes that English high negation only has one reading, (9b). With Russian, one must be vigilant, as negation has only one surface position so it has to be shown that questions like (10) really have high negation, ideally also controlling for intonation. At present, there is no evidence that such questions are inherently biased or that they have the LF in (9c), and not in (9a) (pace G&R, who don’t discuss low negation at all).

To conclude, the exact semantic contribution of high negation in Russian is unclear, and intonation is a confound in both *li*-Qs and Decl-Qs. Furthermore, there is presently no evidence for the outer-inner ambiguity of high negation. But if negation only has two readings, low propositional negation and high outer negation, there is no need for dedicated illocutionary operators, as the bias contribution of negation, if any, can be derived pragmatically (Goodhue 2022b). Before the research outlined in this section becomes available, definite claims on the semantics of negative questions are premature. In particular, there is presently no evidence that such questions have illocutionary operators. In what follows, I will focus exclusively on questions with a positive prejacent.

4 Core data

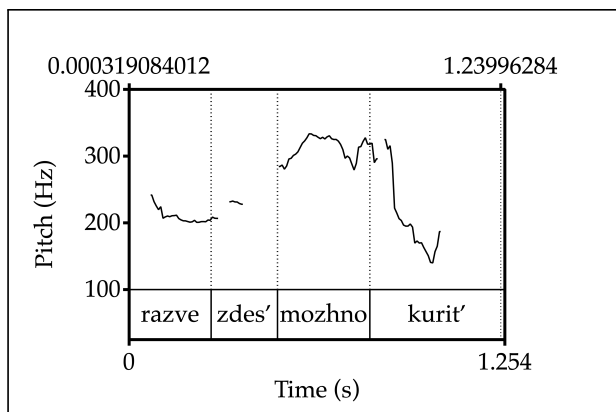
4.1. Prosody and distribution

Razve is often clause-initial, though material can also scramble to its left. *Neuželi* is always clause-initial, a remnant of the historical structure of *neuželi*-Qs as negative *li*-Qs (*neuželi* < *ne* NEG + *uže* ‘already’ + *li*). Both particles can be used stand-alone in responses.

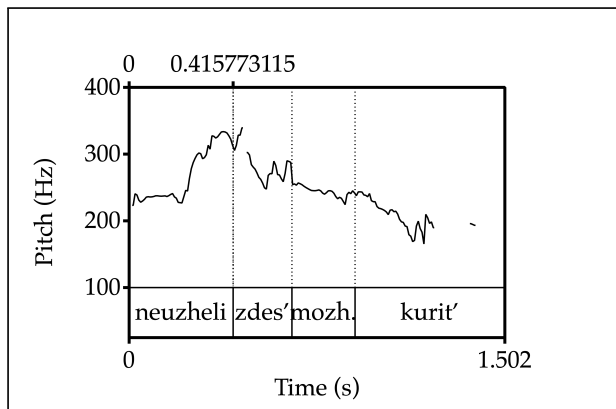
The questions containing the particles differ prosodically, a property not discussed in previous literature. *Razve*-Qs have the same contour and distribution as neutral Decl-Qs:

Q-peak on the main predicate (11) or another constituent, but never clause-finally (they are not compatible with the explanation-seeking contour). *Neuželi*-Qs are different, likely because they started as *li*-Qs diachronically. *Neuželi* itself bears ordinary focus marking (12), same as the fronted constituent of *li*-Qs (5a). This type of prominence is incompatible with the Q-peak in the same clause (cf. especially [Esipova 2025](#) on echo questions), and *neuželi*-Qs never have the Q-peak. In what follows, I will treat focus marking in *neuželi*-Qs as an instance of polarity focus and show in Sections 5.2 and 5.3 that distinct prosodic profiles of *razve*-Qs and *neuželi*-Qs partially determine semantic and conversational contrasts between them.

- (11) **Razve** zdes' možno_Q kurit'?
- RAZVE** here can.PRED smoke.INF
'Can one smoke here?'



- (12) **Neuželi**_F zdes' možno kurit'?
- NEUŽELI** here can.PRED smoke.INF
'Can one smoke here?'



The particles are infelicitous in assertions (they require question intonation), *li*-Qs (13), alternative questions (14), *wh*-questions (15) or embedded polar questions (16). Given that only *li*-Qs (but not Decl-Qs) are licensed under question-embedding predicates, the ungrammaticality of (16) can be due to the incompatibility with *li*, rather than non-embeddability. Crucially, since *li* is not a complementizer, this isn't a syntactic competition between *li* and *razve/neuželi*.¹⁰ *Neuželi*'s incompatibility with *li* is likely due to the fact that it contained *li*

¹⁰In some varieties, *razve* can co-occur with *li* and negation to produce incredulity; this seems to be subject to regional variation. These facts support the idea that (13) is bad due to a semantic, rather than syntactic, clash (Section 5.2) and further show that the meaning of modal particles in general is subject to dialectal

historically, though I will not provide a compositional analysis here.

- (13) **Razve/neuželi** (*li) tancuet (*li) Tata? [**li-Q*]
RAZVE/NEUŽELI LI dance.3SG.PRES LI Tata.NOM
 ‘Does Tata dance? (I thought not)’
- (14) (***razve/neuželi**) Tata tancuet ili net? [**alternative Q*]
RAZVE/NEUŽELI Tata.NOM dance.3SG.PRES or not
 ‘Does Tata dance or not?’
- (15) (***razve/neuželi**) kogda (***razve/neuželi**) Tata tancuet? [**wh-Q*]
RAZVE/NEUŽELI when **RAZVE/NEUŽELI** Tata.NOM dance.3SG.PRES
 ‘When is Tata dancing?’
- (16) *Masha sprosila/uznala/somnevaetsja, [**embedded Q*]
 Masha.NOM ask.F.SG.PST/learn.F.SG.PST/doubt.3SG.PRES
razve/neuželi Tata tancuet.
RAZVE/NEUŽELI T.NOM dance.3SG.PRES
 Intended: ‘Masha asked/learned/doubts if Tata danced.’

4.2. What *razve* does

Razve encodes that the speaker considered *p* unlikely prior to conversation and that there is presently good-enough evidence for *p* (2). This makes belief revision possible but not certain: credence in *p* may have increased due to new evidence, but not enough to accept *p*. Depending on the speaker’s willingness to let go of prior belief, *razve*-Qs have two uses: (i) information-seeking: the speaker is presently unopinionated about *p* (17), and (ii) challenging: the speaker holds on to belief that $\neg p$ (18,19). The latter use falls under the umbrella of what Kiss (2024:112-163) calls ‘adversative questions’, though I will not attempt a direct comparison here. Both uses have the same prosody, non sentence-final Q-peak, and semantics (Korotkova 2023), and only differ in discourse updates (Section 5.3).

In information-seeking *razve*-Qs, the speaker wants to resolve an epistemic conflict (cf. Zalizniak 2020) between prior belief and current evidence (17a). These questions signal no preference towards an answer, and follow-ups that explicitly describe likelihood are infelicitous (17b).¹¹ Pace G&R, *razve* does not correspond to English *really*, which requires a strong present conviction that $\neg p$ (Bill and Koev 2022), thus rendering (17a) borderline rude (*really*-Qs, on the other hand, are predicted to be felicitous with (17b)).

- (17) *I overhear a German friend thanking someone in Greek.*
- a. **Razve** ty govoriš po-grečeski?
RAZVE you:SG.NOM speak.2SG.PRS Greek
 ‘Do you speak Greek? (I thought not.)’
 ≠ ‘Do you really speak Greek?’
- b. Follow-ups to (17a):
 #Čto-to somnevajus’. / #Podozrevaju, čto net.
 PTCL doubt.1SG.PRES suspect.1SG.PRS COMP be.NEG
 Intended: ‘I doubt it’. / ‘I suspect not’.

variation; see recent discussion of German in (Thoma 2016; Zobel 2017, 2025). I thank Masha Esipova for bringing this issue up and leave a detailed investigation for future research.

¹¹The infelicity arises only with predicates encoding ordering about likelihood (see Anand and Hacquard (2013) on *doubt*), while preference-based *hope* and *fear* are fine. This is a correction of (Korotkova 2023:335), where a *doubt*-continuation is deemed acceptable.

In challenging *razve*-Qs (18,19), the speaker protests *p*. Often called ‘rhetorical’ in previous descriptions of *razve* (Rathmayr 1985:117,143-144; Restan 1972:441-445), these uses have an involved pragmatics. Unlike canonical rhetorical questions (Rohde, this volume), they don’t require the answer to be already known by the interlocutors (Korotkova 2023:339-340). And unlike challenging rhetorical questions (Eckardt 2024), they can directly challenge a previous assertion and thus the addressee’s commitment that *p*. They can also be used if the possibility that *p* is raised by a third party (verbally or otherwise) and the speaker hopes to prevent its silent acceptance.

- (18) *Per Chukchi ritual, offerings were left in the snow for the spirits, which a city boy calls ‘buried meat’.*

Razve tak možno nazyvat’ svjaščennuju žertvu?

RAZVE so can.PRED call.INF sacred.FEM.ACC.SG offering.ACC.SG

≈‘How dare you call a sacred offering this way! (Literally: Can one call a sacred offering this way?)’ (The Time of Melting Snows, Yuri Rytkheu)

Common with normative language, challenging uses often convey preferences, but even then the bouletic component is not hard-wired (as noted already by Rathmayr 1985:144).

- (19) illustrates: the speaker is talking about their beliefs, not wishes.

- (19) *Runaway slave George Harris, talking in bitter despair to his former owner, argues that there can’t possibly be a God for Black people.*

Razve Bog est’?

RAZVE God.NOM.SG be.PRES

‘Does God exist? (I don’t think so) (Original ‘Is there a God to trust in?’, with polarity focus on the copula)’

(Uncle’s Tom Cabin, Harriet Beecher Stowe; translated by V.S. Waldman)

4.3. What *neuželi* does

Neuželi is restricted to cases when the speaker considered *p* unlikely and is confronted with evidence that *p* (2). Previous research often classifies *neuželi* (and its colloquial variants *neužel’* and *neužto*)¹² as expressing disbelief: the speaker refuses to accept *p* (Baranov 1986; Bulygina and Shmelev 1997; Repp and Geist 2025). In line with observations by (Rathmayr 1985:146-152) (see also Korotkova 2023), I argue that the particle expresses an opposite doxastic state—the speaker’s upward trending credence in *p*, up to full belief, along with information about discarding prior belief that $\neg p$ based on new evidence. *Neuželi*-Qs have two uses: (i) positively biased: the speaker is leaning towards *p* but isn’t sure (20, 21), and (ii) polar exclamatives: the speaker is astonished that *p* (22,23). Let’s start with positively biased *neuželi*-Qs.

¹²*Neuželi* used to co-exist with the particle *uželi* (< *uže* ‘already’ + *li*), but the extent of their semantic differences remains unclear (Restan 1972:454-455). Now *uželi* is obsolete, appearing exclusively in archaic texts.

(20) *A teenager who has accidentally ended up in the future and unwilling to go back to his time straight away discusses his predicament with a resident robot.*

A. **Neuželi** vy soglasilis' by otpravit'sja v prošloe, tak
NEUŽELI you:PL.NOM agree.PL.PST SUBJ set.off.INF to past.ACC.SG so
 i ne posmotrev na to, što delaetsja v
 and NEG look.CONV at that.N.ACC.SG what.NOM happen.3SG.PRES in
 buduščem?

future.PREP.SG

'Would you agree to go to the past without even having had a look at what's happening in the future? (I thought you wouldn't.)'

B. Da.

yes

'Yes.'

(Film *Guest from the Future*)

(21) *Soviet diplomat Volodin has long been contemplating contacting a foreign ministry about the Soviet Atomic Program. Initially he considers an anonymous call safe but after nerve-wrenching deliberations he is becoming convinced otherwise.*

Neuželi uznajut po telefonnomu sdavlennomu golosu?

NEUŽELI identify.3PL.PRES by phone.M.SG.DAT muffled.M.SG.DAT voice.SG.DAT

'Surely they couldn't identify a muffled voice over a telephone?' (Literally: Will they identify a muffled voice over a telephone?)

(In *The First Circle*, Aleksandr Solzhenitsyn; translated by H. Willets)

In (20), a *neuželi*-Q is used in a dialogue to double-check whether *p* is actually true, even though it was previously considered unlikely. In (21), a *neuželi*-Q in an internal monologue conveys the complexities of a changing epistemic state—from believing $\neg p$ to leaning towards *p*—in a situation with life-or-death consequences, the nuances of the speaker's attitude rendered through the combination of *surely*, negation and a rising declarative in the English literary translation. In each case, the speaker can use the continuation *Ja podozrevaju, čto da* 'I suspect so' (I.NOM suspect.1SG.PRES COMP yes), indicating current high credence in *p*. However, neither case conveys that the speaker is fully certain of the prejacent, so those examples are genuine questions.

Another use of *neuželi*-Qs is what I call exclamative polar questions.¹³ (22), from a scientific space mission, and (23), from a scientific magazine, demonstrate that the speaker now believes *p*, as with the English *can't believe* construction ($\neq$ *don't believe*; Roberts 2019).

(22) *Space ship captain announces that maximum speed has been reached.*

A. Slušaj, **neuželi** my letim so skorost'ju sveta?
 Listen.IMP **NEUŽELI** we.NOM fly.1PL.PRES with speed.ACC.SG light.GEN.SG
 ≈ 'Look, I can't believe we are flying with light speed!'

B. Da, počti.

yes almost

'Yes, almost.'

(Film *Moscow–Cassiopeia*)

¹³This term is a placeholder. Sadock (1974) calls such hybrids 'queclamatives', and questions with miratives have been called 'counterexpectational' (Bhadra 2024) and 'surprise questions' (Celle 2023). However, *neuželi* isn't always mirative. Unlike canonical miratives (Zhuang 2023), it isn't veridical and allows for less-than-full certainty (20,21). In addition to violated expectations (27)—the attitude canonically associated with miratives but in fact different from belief proper (cf. Crespo et al. 2018)—it can also convey discarded background beliefs (26). I leave finessing those points for future research.

(23) *Description of a reader's first encounter with Hawking's multiverse theory.*

Neuželi vokrug našego mira zerkalami rasstavleny
NEUŽELI around our.M.GEN.SG world.GEN.SG mirror.INSTR.PL put.PTCP.PL
 drugie miry?
 other.NOM.PL world.NOM.PL

≈ 'I can't believe that there are other worlds put like mirrors around ours'.

(Magazine *Knowledge is power*)

I argue that *neuželi* has the same core semantics across the board and that there is no reason to treat exclamative uses separately. Unlike standard exclamatives, all *neuželi*-Qs can be answered, including with polarity particles (22,33). And unlike *wh*-exclamatives (Castrviejo 2019; Rett 2011) or verb-initial exclamatives (Brandner 2010), *neuželi*-Qs needn't be degree-based. Like *that*-exclamatives across Germanic (Grosz 2012), they are felicitous with non-gradable properties (23,27) and express a realization that *p*. This suggests that all *neuželi*-clauses are interrogative and that the exclamative effect arises from the level of certainty.¹⁴

4.4. Evidential restrictions

Many natural language expressions, including—but not limited to—grammatical evidentials, convey an evidential signal specified for evidence flavor and/or the strength of belief this evidence produces (Krawczyk 2012; Murray 2017). Yet the literature on questions treats 'contextual bias' as a monolithic notion: mutually available evidence supporting the prejacent (Büring and Gunlogson 2000; Romero 2024; Sudo 2013 a.o.). This view does not anticipate the existence of bias expressions with evidential restrictions. However, as I show below, precisely such restrictions matter for *razve* and *neuželi*.

Evidence flavor Evidential expressions typically lexicalize how the speaker acquired the prejacent, e.g. via perception, inference or hearsay (Willett 1988). Recent research has shown that many such expressions are sensitive to the direction of reasoning, for example, *apparently* (Krawczyk 2012), *will* (Cariani 2022; Cumming and Winans 2021) or *discover* (Korotkova and Anand 2024). Using *razve* as an exemplar, I show that contextual bias exhibits similar restrictions. As I argue in Korotkova (2023), *razve* is best described as requiring an inference from effect (evidence) to its explanation (prejacent), a fact not captured by approaches that simply require the presence of 'contextual bias'.

More specifically, if *p* is the prejacent of the question, there is a prior belief that $\neg p$ and a current observation *q* that constitutes evidence for *p*, *razve*-Qs are felicitous if *p* is a good-fit explanation for *q*¹⁵ (cf. Rathmayr 1985:140) and infelicitous if *q* is an explanation for *p*. *Razve*-Qs are thus compatible with a range of information sources, including sensory evidence (2,3,17,24) and hearsay (4,18), as long as the above schema is in place. Crucially, *razve* is not acceptable when the direction of reasoning is reversed (25).¹⁶

¹⁴NPIs are commonly used to distinguish interrogative vs. exclamative clauses, but this test isn't applicable as Russian lacks canonical NPIs (Pereltsvaig 2006).

¹⁵Such inferences are often called 'abductive' (Douven 2021).

¹⁶One may wonder if the relevant notion here is 'cause', not 'explanation' (assuming that the two are distinct, Lipton 2010). While in some cases we are dealing with causes (24,25), 'explanation' has better empirical coverage overall. Thus, speaking Greek explains thanking in Greek, but doesn't necessarily cause it (17a=26).

(24) Reasoning from effect to explanation

Rain wasn't in the forecast so I was sure it wouldn't rain and put laundry outside. We go outside, laundry is wet, I ask:

Razve/neuželi byl dožd'?

RAZVE/NEUŽELI be.M.SG.PST rain.NOM.SG

razve-Q: 'Did it rain? (I thought it wouldn't.)'

neuželi-Q: 'Oh, so it rained?'

p = 'it rains', q = 'wet laundry', rain explains wet laundry

Neuželi is omnivorous with respect to the direction of reasoning: it is compatible both with inferences from effect to explanation (2,3,22,24,26,27) and explanation to effect (20,21,25). The latter encompasses cases of stereotypical reasoning where the truth of the prejacent is supported not by a specific observation but by general ideas—only *neuželi* is acceptable in those. Thus, in (20) the speaker has no expectations about their interlocutor per se but rather thinks that everyone would behave in a particular way given the circumstances.

(25) Reasoning from explanation to effect

- a. *Venice banned passengers of cruise ships from disembarking on weekdays. It's Monday, a huge ship is stopping.*

✓ **neuželi/#razve** segodnja snova budut tolpy?

NEUŽELI/RAZVE today again be.3PL.FUT crowd.NOM.PL

'Will there will be crowds again today? (I thought there wouldn't be.)'

p = 'crowds', q = 'ships stop', ships explain crowds

(adapted from [Korotkova 2023:337](#))

- b. *Rain wasn't in the forecast so I was sure it wouldn't rain and put laundry outside. An hour later, when we're at work and go for a walk, it is raining. I ask:*

✓ **neuželi/#razve** bel'ë namoklo?

NEUŽELI/RAZVE laundry.NOM.SG get.wet.N.SG.PST

'Oh, did the laundry get wet? (I thought it wouldn't.)'

p = 'wet laundry', q = 'it rains', rain explains wet laundry

Evidence strength *Razve*-Qs and *neuželi*-Qs are associated with different posterior credence in p : (i) low to undecided with *razve*-Qs, (ii) high to full with *neuželi*-Qs. This constrains the subjective strength of contextual evidence. *Razve*-Qs require evidence to be good enough for the speaker to entertain p without accepting it immediately ([Korotkova 2023](#)). *Neuželi*-Qs convey a higher credence in p and therefore typically stronger subjective evidence.

This difference in the epistemic states accounts for the following contrasts. In scenarios that allow both *razve* and *neuželi*, the particles are not interchangeable: in (26) *razve* conveys genuine curiosity about the prejacent, while *neuželi* signals near-certainty. (Note that thanking in Greek doesn't always entail speaking Greek.)

- (26) *I overhear a German friend thanking someone in Greek.*

Razve/neuželi ty govoriš po-grečeski? [=17a]

RAZVE/NEUŽELI you:SG.NOM speak.2SG.PRS Greek

razve-Q: 'Do you speak Greek? (I thought not.)'

neuželi-Q: 'I can't believe you speak Greek!'

When evidence leaves no room for doubt, only *neuželi*-Qs are acceptable.¹⁷ *Razve* is odd in (27), implying that the speaker doesn't trust their eyes. Likewise, a modification of (26) in (28) renders *razve* infelicitous—conversing in Greek unmistakably reveals the ability to speak it.

(27) *Opening the door to someone I never expected to see again.*

✓*neuželi*/#*razve* eto ty?

NEUŽELI/*RAZVE* this you.NOM
'I can't believe this is you!'

(28) *I overhear a German friend having a conversation in Greek.*

a. #*Razve* ty govoriš po-grečeski? 'Do you speak Greek? (I thought not.)'

b. ✓*Neuželi* ty govoriš po-grečeski? 'I can't believe you speak Greek!'

Strength-based asymmetries also manifest in dialogues. When the prejacent is a previous assertion, *razve*-Qs are acceptable in their challenging function (18,19), but are impolite otherwise (Bulygina and Shmelev 1997:275-276). Thus, *razve* will be odd in (22), suggesting that the captain's announcement that *p* is insufficient to believe *p*. *Neuželi*-Qs, on the other hand, are natural in those scenarios, as they convey (near-)certainty.

Both *razve*-Qs and *neuželi*-Qs encode that the speaker considered *p* unlikely and that there is contextual evidence supporting *p*. They differ in what they convey about the speaker's current epistemic state and contextual evidence, these subtle distinctions not predicted by the existing typology: (i) with *razve*-Qs, the speaker may be undecided about *p* (information-seeking uses) or still believe $\neg p$ (challenging uses), and contextual evidence is plausibly explained by *p*; (ii) with *neuželi*-Qs, the speaker's credence in *p* has increased (opinionated uses) or they may have accepted it (exclamative uses), and contextual evidence strongly suggests *p*. I analyze the particles as pure doxastic operators and argue that their discourse effects are rooted in this basic semantics.

5 Proposal

5.1. Polar interrogatives

I assume a type distinction between declaratives and interrogatives (propositions vs. sets of propositions) and the monopolar semantics for polar questions (singleton sets; the idea goes back to Bolinger 1978), with questionhood encoded by the Q operator in C (29; adapted from Biezma and Rawlins 2012:392). Note that the prejacent of a question must be one of the salient alternatives—a question must address the QUD.

(29) $\llbracket [\text{CP } Q [p]] \rrbracket^c = \{\lambda w.p(w)\},$
defined iff (i) $\{\lambda p.p(w)\} \subseteq \text{ALTS}(c)$ or $\text{ALTS}(c) = \emptyset$, and
(ii) $|\{\lambda p.p(w)\} \cup \text{ALTS}(c)| > 1.$

(ALTS(*c*) is a set of salient propositions that are possible answers to the QUD.)

For Russian *li*-Qs and Decl-Qs, I assume the following properties: (i) neither *li* nor the Q-peak encodes the Q operator directly (questions not marked in any way are ruled out

¹⁷It is about evidence strength, not type, as the two needn't be correlated—e.g., perception doesn't always produce more certainty than inference (Faller 2001; McCready 2015; von Stechow and Gillies 2010).

by the pressure to distinguish questions from assertions), (ii) *li* is a binary alternative operator that can combine with questions, (iii) the Q-peak selects for the Q operator, which limits it to questions, and adds the use-conditional meaning “Please react” (Esipova 2024). We can now define the semantics for *li*-Qs (30; with the contribution of focus intonation suppressed) and Decl-Qs (31).¹⁸

- (30) a. *Možno li zdes’ kurit’?* ‘Can one smoke here?’ [li-Q; =5a]
 b. $\llbracket (30a) \rrbracket^c = \llbracket [_{CP} li [Q [one\ can\ smoke\ here]]] \rrbracket^c$
 $= \{CAN\ SMOKE, \neg CAN\ SMOKE\}$,
 defined iff (i) $\{CAN\ SMOKE\} \subseteq ALTS(c)$ or $ALTS(c) = \emptyset$, and
 (ii) $|\{CAN\ SMOKE\} \cup ALTS(c)| = 2$.
- (31) a. *Zdes’ možno kurit’?* ‘Can one smoke here?’ [Decl-Q; =5b]
 b. $\llbracket (31a) \rrbracket^c = \llbracket [_{CP} Q\text{-peak} [Q [one\ can\ smoke\ here]]] \rrbracket^c$
 $= \{CAN\ SMOKE\}$,
 defined iff (i) $\{CAN\ SMOKE\} \subseteq ALTS(c)$ or $ALTS(c) = \emptyset$, and
 (ii) $|\{CAN\ SMOKE\} \cup ALTS(c)| > 1$;
 if defined, felicitous iff (iii) Speaker wants Addressee to react.

Any standard semantics predicts that the interpretation of a *li*-Q with *li* on the main predicate contains only two alternatives. Crucially, the strictly doubleton semantics (30) is needed to account for cases when *li*’s host is a non main predicate constituent. Even then, unlike in corresponding Decl-Qs with the Q-peak on a non main predicate constituent, *li*-Qs are limited to scenarios with two salient alternatives (32), which shows that this property must be hard-wired.

- (32) ✓**Two alternatives:** *Dima plans to go either to Berlin or somewhere else.*
 #**More alternatives:** *Dima plans to visit a European city: Amsterdam, Berlin, Copenhagen.*
 V Berlin li sobiraetsja poexat’ Dima?
 in Berlin.ACC LI intend.3SG.PRES go.INF Dima.NOM
 ‘Is it Berlin that Dima is going to visit?’
 (adapted from an earlier version of Shushurin 2024)

5.2. Core semantics

Our first stop is the semantic status of speaker bias and contextual bias, both of which I analyze as not-at-issue entailments (cf. Xu 2017). First, neither *razve*-Qs nor *neuželi*-Qs can be followed up with *I never considered this issue* (cancelling prior belief) or *We have no reasons to think so* (cancelling evidence), which shows that each inference is hard-wired. Second, responses can only target the prejacent (33), but not the speaker’s evidence or prior belief, which shows that only the prejacent is part of the at-issue content (Amaral et al. 2007).¹⁹

- (33) A. **Razve/neuželi** zdes’ možno kurit’?
RAZVE/NEUŽELI here can.PRED smoke.INF
 ‘Can one smoke here? (I thought not.)’

¹⁸The Q-peak also appears in ‘concerned’ requests, including with verbs in the imperative (Esipova 2024). I assume that those are secondary speech acts that originate as interrogatives.

¹⁹Due to the restricted distribution of the particles, other diagnostics of (not-)at-issueness do not apply.

B. Da. ‘Yes’.	B’. Net. ‘No’.
= ‘One can smoke.’	= ‘One can’t smoke.’
≠ ‘You thought so’.	≠ ‘You didn’t think so.’
≠ ‘There is evidence for this.’	≠ ‘There is no evidence for this.’

Both biases are conventionally encoded by *razve/neuželi* and constitute a type of not-at-issue content, though of different sorts. The proposition that is considered evidence is part of mutual knowledge, but the speaker’s attitude can be new information. (34) illustrates (as discussed in [Korotkova \(2023\)](#), the mutual evidence requirement is easy to accommodate).

(34) *We’re in a bar. I think smoking is universally banned and don’t bring it up beforehand.*

✓**Mutual evidence:** We watch someone light a cigarette.

#**Non-mutual evidence:** You were at the counter, someone lit a cigarette.

Razve/neuželi zdes’ možno kurit’? ‘Can one smoke here?’ (=33)

Recall from Section 4.1 that *razve* and *neuželi* differ in the question strategies they combine with: Decl-Qs for *razve*-Qs, *li*-Qs (at least historically) for *neuželi*-Qs. Importantly, the bias restrictions discussed in this paper are contributed by the particles themselves rather than those question strategies. Both Decl-Qs (with non clause final Q-peak) and *li*-Qs are neutral in terms of bias ([Esipova and Korotkova 2023](#)): they don’t require the speaker to have any prior beliefs about the likelihood of the prejacent, they don’t hard-wire the speaker’s undecidedness and can be used in quiz questions, and they don’t require the presence of evidence supporting the prejacent (though, as is typical for polar questions with positive prejacent in general, they are incompatible with evidence contradicting the prejacent; cf. footnote 7). That said, some restrictions on *razve*-Qs and *neuželi*-Qs have to do with the underlying question strategies.

Razve and *neuželi* have a limited distribution (Section 4.1). They are out in assertions, embedded questions, *li*-Qs, alternative questions and *wh*-questions. Such restrictions seem to be common for biased questions across languages, however, I propose that they stem from different sources in each case. The distribution of *razve*-Qs is the same as that of Decl-Qs they are based of: the Q-peak, obligatory in *razve*-Qs, is generally incompatible with the ordinary focus marking always present in *wh*-questions, *li*-Qs and alternative questions ([Esipova 2024](#); [Meyer and Mleinek 2006](#)). Following [Korotkova \(2023\)](#) (inspired by [Bhatt and Dayal 2020](#)), I argue that *razve* selects for sets of propositions (35), which limits it to questions, and imposes a singleton constraint, which limits it to Decl-Qs and excludes *li*-Qs.²⁰ First, *wh*-questions, alternative questions and *li*-Qs are not singletons. Second, the particle’s incompatibility with *li* (the only strategy for embedded polar questions across the board; (6)) ensures that it is limited to matrix clauses without any additional assumptions about embeddability or the denotation of embedded questions. We can now provide compositional semantics for *razve* (35).²¹ I do not delineate a proper logical account of belief

²⁰The inquisitive-semantics notion ‘highlighting’ ([Roelofsen and Farkas 2015](#)) can also be used to explain the distribution (cf. [Xu 2017](#) on *nandao*). We can treat *razve* and *neuželi* as selecting for clauses that highlight only one alternative, which predicts their unacceptability in *wh*-questions and alternative questions (which highlight more than one) and *li*-Qs (if those are treated as highlighting exactly two alternatives, cf. (32)).

²¹Caveats about the implementation in (35) and (36). A. The presence of salient information imposes a constraint on the input context, treated as a definedness condition. B. The speaker’s attitude to *q* needn’t be mutual knowledge and is modelled as a novel use-conditional contribution (same as the speaker’s credence). A complete model of different not-at-issue contributions is beyond the scope of this paper, but the important point is that this inference is not presuppositional in the classic sense. C. The probabilistic treatment of evidence is standard in epistemology ([Williamson 2000](#)); see [Davis et al. \(2007\)](#); [McCready and Ogata \(2007\)](#);

change in response to new evidence, which can be done in a number of ways, and just assume that each operator lexically encodes information about credence and evidence (cf. footnote 6). I hope to see refinements on this in future work.

- (35) $\llbracket \text{razve} \rrbracket^c = \lambda P_{\langle\langle st \rangle t \rangle} . P$
- a. Defined iff $|P| = 1$ and $\exists q \in cg$;
 - b. If defined, felicitous iff $\forall p \in P$

$[c_t(p | q) > c_t(p),$
and p explains $q,$
and $\exists t'. [t' < t \wedge 0 \leq c_{t'}(p) \leq 0.25],$
and $0 \leq c_t(p) \leq 0.5].$

$[q \text{ is evidence for } p]$
 $[\text{direction of reasoning}]$
 $[\text{prior credence}]$
 $[\text{posterior credence}]$

(where cg is the common ground, c —credence, t —the utterance time.)

For *neuželi*-Qs, I propose that their limited distribution has to do with the type of focus prominence they have, though I cannot argue for this in detail for reasons of space. Recall that *neuželi* bears ordinary focus marking (12). Given that *neuželi*-Qs are polar questions that can be felicitously answered with polar particles (20, 22), this is an instance of polarity focus, which, in turn, is generally incompatible with *wh*-Qs and alternative questions due to a clash in the alternatives they generate. As for the incompatibility with *li*, we're likely dealing with a mix of synchronic and diachronic factors. *Neuželi* can be traced back to negative *li*-Qs, so it cannot combine with *li* since it already contains it, though only a more detailed future investigation can show to what extent this *li* is active semantically and/or syntactically. The compositional semantics is given below (36), with the exact contribution of focus suppressed for simplicity. Like *razve*, *neuželi* selects for sets of propositions, which limits it to questions, but doesn't impose a singleton constraint on its prejacent.

- (36) $\llbracket \text{neuželi} \rrbracket^c = \lambda P_{\langle\langle st \rangle t \rangle} . P$
- a. Defined iff $\exists q \in cg$;
 - b. If defined, felicitous iff $\forall p \in P$

$[c_t(p | q) > c_t(p),$
and $\exists t'. [t' < t \wedge 0 \leq c_{t'}(p) \leq 0.25],$
and $0.75 \leq c_t(p) \leq 1].$

$[q \text{ is evidence for } p]$
 $[\text{prior credence}]$
 $[\text{posterior credence}]$

(where cg is the common ground, c —credence, t —the utterance time.)

In words: *razve* and *neuželi* select for sets of propositions. Questions containing the particles signal that (i) the speaker considered the prejacent unlikely prior to conversation, (ii) there is mutually known salient information q , and (iii) the speaker considers it to be evidence for the prejacent (q is a good predictor of p). One of the key differences between *razve*-Qs and *neuželi*-Qs is posterior credence.²² With *razve*-Qs, there is an effect-to-explanation inference that p and the speaker may consider p unlikely or be genuinely undecided. With *neuželi*-Qs, the speaker is (nearly) certain that p . Note that the strength of evidence isn't hard-wired in either case—it is derived from prior and posterior credences. A notational alternative would be to hard-wire evidence strength and derive posterior cre-

McCready (2015) on evidentials. D. Conditional probabilities are insufficient for causes (Pearl 2009) or explanations, so the specific evidence relevant for *razve* is unformalized for simplicity. E. Probability thresholds reflect likelihood: unlikely below 25%, likely above 75%.

²²G&R posit that the particles differ in the strength of epistemic conflict between prior belief and new evidence. A direct comparison is difficult given that they do not provide a formal analysis, but as far as I understand it, my view is compatible with this idea (posterior = prior conditionalized on evidence).

dence. Finally, if *neuželi*-Qs have polarity focus, the speaker's present credence in the pre-jacent can potentially be attributed to that rather than the particle itself, given that such focus is frequently associated with certainty and has often been viewed as an instance of the illocutionary operator VERUM (see detailed discussion in Goodhue (2022a)). I remain open to a more compositional analysis of the meaning of *neuželi*-Qs when more research becomes available.

5.3. Conversational moves

We now derive the full meaning of *razve*-Qs and *neuželi*-Qs, including their conversational effects. I start with *razve*-Qs.

(37) *We're in what we thought was a non-smoking bar. Someone lights a cigarette.*

Razve zdes' možno kurit'? [=11]

RAZVE here can.PRED smoke.INF

'Can one smoke here?'

(38) $\llbracket (37) \rrbracket = \llbracket \text{razve} [\text{CP Q-peak} [\text{Q} [\text{one can smoke here}]]] \rrbracket^c = \{\text{CAN SMOKE}\}$

- a. Defined iff $\{\text{CAN SMOKE}\} \subseteq \text{ALTS}(c)$ or $\text{ALTS}(c) = \emptyset$,
and $|\{\text{CAN SMOKE}\}| = 1$,
and $\exists q \in cg$ ($q = \text{LIT CIGARETTE}$);
- b. If defined, felicitous iff Speaker wants Addressee to react,
and $c_t(\text{CAN SMOKE} \mid \text{LIT CIGARETTE}) > c_t(\text{CAN SMOKE})$,
and CAN SMOKE explains LIT CIGARETTE,
and $\exists t'. [t' < t \wedge 0 \leq c_{t'}(\text{CAN SMOKE}) \leq 0.25]$,
and $0 \leq c_t(\text{CAN SMOKE}) \leq 0.5$.

(where cg is the common ground, c —credence, t —the utterance time.)

Razve-Qs have two uses (section 4.2). I argue that the actual interpretation in context depends on the speaker's current credence in p :

- information-seeking: speaker undecided, $\approx 0.25 \leq C_t(p) \leq 0.5$;
- challenging: speaker unwilling to abandon prior belief, $\approx 0 \leq C_t(p) \leq 0.25$.

I represent the contrast between different interpretations via the Table model (Farkas and Bruce 2010), a model of discourse that consists of the sets DC_{Sp} and DC_{Ad} for individual discourse commitments of the interlocutors, the Table for issues under discussion, the common ground cg and the projected set ps for possible outcomes of a discourse move.²³ I assume the following architecture of discourse: (i) assertions and questions are associated with different updates;²⁴ (ii) questions put the set of salient focus alternatives on the Table, which amounts to $\{p, \neg p\}$ for polar questions; (iii) only at-issue content is represented; (iv) discourse commitments are all-or-nothing, not graded (cf. Rudin 2022). (39) shows the update of a neutral information-seeking question in (31a).

(39) Information-seeking question

Zdes' možno kurit'? 'Can one smoke here?'

²³Nothing hinges on this choice, and the same intuition can be expressed in other frameworks modelling discourse structure, e.g., commitment-space semantics (Krifka 2015, 2021).

²⁴One could use the general update function (Farkas and Roelofs 2017) and treat declaratives as sets, but then the incompatibility of *razve* and *neuželi* with *li*-Qs remains unexplained (cf. footnote 20).

K_0	DC_{Sp}	Table	DC_{Ad}	$\rightarrow K_1$	DC_{Sp}	Table	DC_{Ad}
						$\{p, \neg p\}$	
	$c g_0$				$c g_1 = c g_0$	$ps_1 = \{c g_1 \cup \{p\}, c g_1 \cup \{\neg p\}\}$	

Turning back to *razve*, I assume the following: (i) all *razve*-clauses are interrogative; (ii) credence determines commitment; (iii) *razve* introduces a constraint on the input CG. I argue that in those cases when the speaker is genuinely undecided at present (2,4a,17a=26,24), the discourse effect is only minimally different from ordinary information seeking questions. (40) illustrates this for (37).

(40) Information-seeking *razve*-Q: undecided about p

Razve zdes' možno kurit'? 'Can one smoke here?'

$\approx$ I thought one couldn't, but now I see it could be allowed. So is it allowed or not?

K_0	DC_{Sp}	Table	DC_{Ad}	$\rightarrow K_1$	DC_{Sp}	Table	DC_{Ad}
						$\{p, \neg p\}$	
	$q \in c g_0$				$c g_1 = c g_0$	$ps_1 = \{c g_1 \cup \{p\}, c g_1 \cup \{\neg p\}\}$	

Recall from 4.2 that challenging *razve*-Qs usually convey the speaker's protest against p and its acceptance. (37) allows this interpretation, for example, if the speaker is adamant that smoking is evil and would love to extract a similar commitment from interlocutors. I argue that this interpretation arises when the speaker considered p unlikely and still does (41).

(41) Challenging *razve*-Qs: conviction that $\neg p$

Razve zdes' možno kurit'? 'Can one smoke here?'

$\approx$ Someone is smoking, but it should be banned, let's agree that it should.

K_0	DC_{Sp}	Table	DC_{Ad}	$\rightarrow K_1$	DC_{Sp}	Table	DC_{Ad}
		$\{p\}$	optional: $\{p\}$		$\{\neg p\}$	$\{p, \neg p\}$	optional: $\{p\}$
	$q \in c g_0$	$ps = \{c g_0 \cup \{p\}\}$			$c g_1 = c g_0$	$ps_1 = \{c g_1 \cup \{p\}, c g_1 \cup \{\neg p\}\}$	

(41) is a non-monotonic update: instead of decreasing the space of live options, it increases them. It may require the addressee to revise commitments, which is an adversarial discourse move (Kiss 2024; Rudin 2022). This is a welcome result, as it reflects the discourse dynamics associated with challenging questions—they challenge—and makes good on the intuition that some *razve*-Qs are impolite, especially when the preface of such question is a previous assertion.

We now turn to *neuželi* and its discourse effects.

(42) *We're in what we thought was a non-smoking bar. Someone lights a cigarette.*

neuželi zdes' možno kurit'? [=12]

NEUŽELI here can.PRED smoke.INF

'Can one smoke here?'

(43) $\llbracket (42) \rrbracket = \llbracket \text{neueli} [_{CP} \text{Q-peak} [\text{Q} [\text{one can smoke here}]]] \rrbracket^c = \{\text{CAN SMOKE}\}$

- Defined iff $\{\text{CAN SMOKE}\} \subseteq \text{ALTS}(c)$ or $\text{ALTS}(c) = \emptyset$,
and $\exists q \in c g$ ($q = \text{LIT CIGARETTE}$);
- If defined, felicitous iff Speaker wants Addressee to react,
and $c_t(\text{CAN SMOKE} \mid \text{LIT CIGARETTE}) > c_t(\text{CAN SMOKE})$,

and $\exists t'. [t' < t \wedge 0 \leq c_{t'}(\text{CAN SMOKE}) \leq 0.25]$,
and $0.75 \leq c_t(\text{CAN SMOKE}) \leq 1$.

(where c_g is the common ground, c —credence, t —the utterance time.)

Neuželi-Qs have two uses (section 4.3), and as with *razve*, those differences depend on the speaker's credence:²⁵

- positively biased: leaning towards p , $\approx 0.75 \leq c_t(p) < 1$;
- exclamative: astonished that p , $\approx c_t(p) = 1$.

The Table model is not well-suited for exclamatives (unless not-at-issue content, including the expressive content of exclamatives, is incorporated; cf. Scheffler and Malamud 2023). Thus, (44) collapses the two uses of *neuželi*-Qs, and only models the distinction between all *neuželi*-Qs vs. ordinary information-seeking questions via the speaker's commitment to p .²⁶

(44) ***Neuželi*-Qs**

Neuželi zdes' možno kurit'? 'Oh, one can smoke here?'

Biased interpretation, opinionated speaker: Can you confirm?

Exclamative interpretation, astonished speaker: This isn't what I thought.

K_0	DC _{Sp}	Table	DC _{Ad}	$\rightarrow K_1$	DC _{Sp}	Table	DC _{Ad}
					{ p }	{ $p; \neg p$ }	
	$q \in c g_0$				$c g_1 = c g_0$	$p s_1 = \{c g_1 \cup \{p\}, c g_1 \cup \{\neg p\}\}$	

The pragmatics I propose for *razve*-Qs (40,41) and *neuželi*-Qs ([*neuželi*-update]) follows naturally from the compositional semantics of the particles. Each particle encodes credence in the prejacent: low to undecided for *razve*, high to full for *neuželi*. My key assumption is that credence determines the level of commitment, which determines the range of conversational moves available to each particle. Each particle is also compatible with a wide spectrum of emotions, but the semantic requirements only concern credence and evidence.

By treating bias-inducing expressions as doxastic operators, the proposed account shifts the locus of variation into semantics. Many expressions convey various degrees of credence, and they may create bias when used in questions (e.g., weak epistemic adverbials; Krifka 2024). Crucially, those discourse effects needn't be treated as primitives, but can be derived in combination with standard assumptions about the pragmatics of questions, in line with other approaches that share the same spirit, if not implementation (Bill and Koev 2022; Goodhue 2022b; Rudin 2022; Silk 2019).

Finally, there is another discourse difference between *razve*-Qs and *neuželi*-Qs that is rooted in the differences between the types of interrogatives they can combine with. As discussed in (Esipova and Korotkova 2023), only *li*-Qs, but not Decl-Qs, can be used in

²⁵*neuželi* is also used sarcastically (especially with cues like breathy voice), which is derivable from its core semantics via hyperbole (cf. Nouwen 2024): an expression of high certainty conveys doubt via scale reversal. *Razve*, precisely because it lacks certainty, has no such uses.

²⁶Maybe polar exclamative questions are assertions: the speaker commits to p and puts it on the Table. But then we sacrifice the idea that all *neuželi*-clauses are interrogative. It is possible that exclamative *neuželi*-Qs differ from questions in additional presuppositions (cf. Zanuttini and Portner 2003 on *wh*-exclamatives). For now, I assume that the Table model must not account for all possible discourse effects, especially those outside of assertions and polar questions.

conjectural questions (of the type discussed by Eckardt, this volume). Such questions constitute non-obtrusive, and often self-directed, inquiries that can be paraphrased as *I wonder whether ϕ* . This lack of pressure to respond is incompatible with the Q-peak (which aims at eliciting a reaction from the addressee) and as such with Decl-Qs. Recall from Section 4.1 that *razve*-Qs require the Q-peak while *neuželi*-Qs never have it. Due to this difference, only *neuželi*-Qs are felicitous as self-directed (21) and other types of conjectural questions, while *razve*-Qs aren't. This contrast was noted already by Rathmayr (1985:145), albeit without a discussion of the intonation that is ultimately responsible for it.

5.4. On operator-based approaches

It is common to analyze question bias via illocutionary operators VERUM (Gutzmann and Castroviejo Miró 2011; Romero and Han 2004), conveying the speaker's desire for p to become common ground, and FALSUM (Frana and Rawlins 2019; Repp 2013), conveying the speaker's protest against p 's becoming common ground.

For Russian, G&R hypothesize that *neuželi* instantiates VERUM while *razve* is an operator co-occurring with VERUM or FALSUM. First of all, G&R only examine negative questions and assume that high negation is ambiguous between the outer, FALSUM reading ([Q [FALSUM [p]]]; (9b)), and the inner, VERUM reading ([Q [VERUM [$\neg p$]]]; (9b)). Recall from 3.3 that there is no evidence for this ambiguity in Russian, which undermines the rationale for postulating the operators. The account also faces additional problems:

- if bias is always produced by operators supplied by negation, it is unclear how to derive bias in *razve*-Qs with a positive prejacent (unless *razve* itself is taken to express one of the operators);
- if *razve* always co-occurs with VERUM or FALSUM, *razve*-Qs are incorrectly predicted to be infelicitous if the speaker is genuinely undecided about p , contrary to fact (17). Even though the operators don't encode the speaker's doxastic state directly, they convey the speaker's certainty as to which propositions should become common ground (Frana and Rawlins 2019; Romero and Han 2004). This, in turn, leads to the speaker's active preference for an answer, which is not a correct description of *razve*-questions (Korotkova 2023).²⁷
- *neuželi* itself is too complex for VERUM: typical markers of VERUM simply express the speaker's certainty in light of possible controversy and aren't limited to polar questions (Jordanoska et al. 2023), while *neuželi*-Qs express certainty after having discarded the opposite belief in light of new evidence. Furthermore, it is important to better understand the prosody of *neuželi*-Qs. It is likely that VERUM-like effects observed by G&R are due not to the particle but to the obligatory presence of polarity focus, which in turn can be derived from a general theory of focus without postulating illocutionary operators (Goodhue 2022a).²⁸

To sum up, there is currently no evidence that Russian questions—with or without negation—necessitate an analysis with VERUM or FALSUM. Absent such evidence, the con-

²⁷The pattern in (17) is replicated with negation, *Razve ty ne zabył grecheskij?* 'Did you not forget Greek? (I thought you did)'. The speaker is undecided, which can be analyzed as *razve* + low propositional negation, no operators needed.

²⁸A reviewer asks: how can an operator-less proposal explain why negative *neuželi*-Qs don't license the PPI *uzhe* 'already' (noted by G&R), given that it seems to require high negation (Zanon 2025).). As stressed throughout the paper, we first need to understand the meaning and intonation of negative questions without any particles—some of them might be clashing with *neuželi*.

tribution of *razve* and *neuželi* can also be analyzed without those operators, which is the route I advocate in this paper.

6 Outlook

Discourse particles are small words that explicitly bring out often hidden aspects of conversational information flow (Grosz 2020; McCready 2012; Zimmerman 2011), and they have played an important role in elucidating the subtleties of question meaning across languages (see, for example, Bayer, Bhadra, Bhatt et al, Chang, Faller this volume). Focusing on two question particles in Russian, *razve* and *neuželi*, this paper offers a new perspective on the phenomenon of question bias.

The combinatorics of different bias types predicts simultaneously less and more variation than we find. On the one hand, current typology does not predict fine-grained differences between *razve* and *neuželi*, which track changes in the speaker’s doxastic state in light of new evidence. On the other, it incorrectly predicts the existence of expressions that only convey negative speaker bias, without evidential restrictions. I have argued that belief revision provides an instrumental way of understanding ‘negative speaker bias’, capturing the behavior of *razve* and *neuželi* in a principled way. I have also argued that ‘contextual bias’ is best understood through the prism of research on evidence in language, as it has restrictions found with evidential expressions (construed very broadly; see overview in Bary and Korotkova 2023).

Needless to say, many open questions remain. This is one of the early forays into biased questions in Russian and much needs to be done, including discerning the roles of prosody and negation in different question types. Then, a careful investigation of the types of contextual bias with an eye on evidential distinctions is in order. And finally, the conceptualization of bias through belief revision needs to be tested cross-linguistically (cf. especially extensive discussion in Kiss 2024). I implore the reader to further advance, challenge, endorse, and sharpen this research agenda.

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